# 2 Matrices




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## 2.1 Operations with Matrices



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## Objectives

- Determine whether two matrices are equal.
- Add and subtract matrices and multiply a matrix by a scalar.
- Multiply two matrices.
- Use matrices to solve a system of linear equations.
- Partition a matrix and write a linear combination of column vectors.

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## Equality of Matrices

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## Equality of Matrices (1 of 2)

**Definition of Equality of Matrices**

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are **equal** when they have the same size ($m \times n$) and $a_{ij} = b_{ij}$ for $1 \le i \le m$ and $1 \le j \le n$.

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## Example 1 – Equality of Matrices

Consider the four matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad C = [1 \quad 3], \quad \text{and} \quad D = \begin{bmatrix} 1 & 2 \\ x & 4 \end{bmatrix}.$$

Matrices $A$ and $B$ are **not** equal because they are of different sizes. Similarly, $B$ and $C$ are not equal.

Matrices $A$ and $D$ are equal if and only if $x = 3$.

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## Equality of Matrices (2 of 2)

A matrix that has only one column, is a **column matrix** or **column vector.**

Similarly, a matrix that has only one row is a **row matrix** or **row vector.**

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## Matrix Addition, Subtraction, and Scalar Multiplication

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## Matrix Addition, Subtraction, and Scalar Multiplication (1 of 2)

To **add** two matrices (of the same size), add their corresponding entries.

**Definition of Matrix Addition**

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrices of size $m \times n$, then their **sum** is the $m \times n$ matrix $A + B = [a_{ij} + b_{ij}]$.

The sum of two matrices of different sizes is undefined.





## Example 2 – Addition of Matrices

**a.** $\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -1+1 & 2+3 \\ 0+(-1) & 1+2 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix}$

**b.** $\begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix}$

**c.** $\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

**d.** $\begin{bmatrix} 2 & 1 & 0 \\ 4 & 0 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix}$ is undefined.

## Matrix Addition, Subtraction, and Scalar Multiplication (2 of 2)

**Definition of Scalar Multiplication**

If $A = [a_{ij}]$ is an $m \times n$ matrix and $c$ is a scalar, then the **scalar multiple** of $A$ by $c$ is the $m \times n$ matrix $cA = [ca_{ij}]$.

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## Example 3 – Scalar Multiplication and Matrix Subtraction

For the matrices $A$ and $B$, find (a) $3A$, (b) $-B$, and (c) $3A - B$.

$$A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix}$$

**Solution:**

$$\textbf{a.}\ 3A = 3\begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 3(1) & 3(2) & 3(4) \\ 3(-3) & 3(0) & 3(-1) \\ 3(2) & 3(1) & 3(2) \end{bmatrix} = \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix}$$

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## Example 3 – Solution

**b.** $-B = (-1)\begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ -1 & 4 & -3 \\ 1 & -3 & -2 \end{bmatrix}$

**c.** $3A - B = \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 12 \\ -10 & 4 & -6 \\ 7 & 0 & 4 \end{bmatrix}$

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## Matrix Multiplication

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# Matrix Multiplication (1 of 1)

**Definition of Matrix Multiplication**

If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix, then the **product**

$AB$ is an $m \times p$ matrix

$$AB = [c_{ij}]$$

where

$$\begin{aligned} c_{ij} &= \sum_{k=1}^{n} a_{ik} b_{kj} \\ &= a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj}. \end{aligned}$$

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# Example 4 – Finding the product of two Matrices

Find the product $AB$, where

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}.$$

**Solution:**

First, note that the product $AB$ is defined because $A$ has size $3 \times 2$ and $B$ has size $2 \times 2$. Moreover, the product $AB$ has size $3 \times 2$, and will take the form

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

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## Example 4 – Solution (1 of 3)

To find $c_{11}$, multiply corresponding entries in the first row of $A$ and the first column of $B$. That is,

$$c_{11} = (-1)(-3) + (3)(-4) = -9$$

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}\begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

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## Example 4 – Solution (2 of 3)

Similarly, to find $c_{12}$, multiply corresponding entries in the first row of $A$ and the second column of $B$ to obtain

$$c_{12} = (-1)(2) + (3)(1) = 1$$

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}\begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

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## Example 4 – Solution (3 of 3)

Continuing this pattern produces the results shown below.

$$\begin{aligned}
c_{21} &= (4)(-3) + (-2)(-4) &&= -4 \\
c_{22} &= (4)(2) + (-2)(1) &&= 6 \\
c_{31} &= (5)(-3) + (0)(-4) &&= -15 \\
c_{32} &= (5)(2) + (0)(1) &&= 10
\end{aligned}$$

The product is

$$AB = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix}.$$





## Systems of Linear Equations

## Systems of Linear Equations (1 of 1)

One practical application of matrix multiplication is representing a system of linear equations. Note how the system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

can be written as the matrix equation $A\mathbf{x} = \mathbf{b}$, where $A$ is the coefficient matrix of the system, and $\mathbf{x}$ and $\mathbf{b}$ are column matrices.

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{x}} \underset{=}{=} \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}_{\mathbf{b}}$$



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## Example 6 – Solving a System of Linear Equations

Solve the matrix equation $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 3 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \text{and} \quad \mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

**Solution:**

As a system of linear equations, $A\mathbf{x} = \mathbf{0}$ is

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_1 + 3x_2 - 2x_3 &= 0. \end{aligned}$$



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# Example 6 – Solution (1 of 3)

Using Gauss-Jordan elimination on the augmented matrix of this system, you obtain

$$\begin{bmatrix} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & -\frac{4}{7} & 0 \end{bmatrix}.$$

So, the system has infinitely many solutions. Here a convenient choice of a parameter is $x_3 = 7t$, and you can write the solution set as

$x_1 = t$, $x_2 = 4t$, $x_3 = 7t$, $t$ is any real number.





# Example 6 – Solution (2 of 3)

In matrix terminology, you have found that the matrix equation

$$\begin{bmatrix} 1 & -2 & 1 \\ 2 & 3 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

has infinitely many solutions represented by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ 4t \\ 7t \end{bmatrix} = t\begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}, \quad t \text{ is any scalar.}$$

## Example 6 – Solution (3 of 3)

That is, any scalar multiple of the column matrix on the right is a solution.

Here are some sample solutions:

$$\begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \\ 14 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ and } \begin{bmatrix} -1 \\ -4 \\ -7 \end{bmatrix}.$$





## Partitioned Matrices

# Partitioned Matrices (1 of 2)

The system $A\mathbf{x} = \mathbf{b}$ can be represented in a more convenient way by partitioning the matrices $A$ and $\mathbf{x}$ in the manner shown below. If

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

are the coefficient matrix, the column matrix of unknowns, and the right-hand side, respectively, of the $m \times n$ linear system $A\mathbf{x} = \mathbf{b}$.

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# Partitioned Matrices (2 of 2)

Then

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{b}$$

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# Linear Combinations of Column Vectors



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## Linear Combinations of Column Vectors (1 of 2)

The matrix product $A\mathbf{x}$ is a linear combination of the column vectors $\mathbf{a}_1$, $\mathbf{a}_2$, ..., $\mathbf{a}_n$ that form the coefficient matrix $A$.

$$x_1\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2\begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n\begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$



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## Linear Combinations of Column Vectors (2 of 2)

Furthermore, the system

$A\mathbf{x} = \mathbf{b}$

is consistent if and only if **b** can be expressed as such a linear combination, where the coefficients of the linear combination are a solution of the system.

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## Example 7 – Solving a System of linear Equations (1 of 2)

The linear system

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0 \\ 4x_1 + 5x_2 + 6x_3 &= 3 \\ 7x_1 + 8x_2 + 9x_3 &= 6 \end{aligned}$$

can be rewritten as a matrix equation $A\mathbf{x} = \mathbf{b}$, as shown below.

$$x_1\begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + x_2\begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + x_3\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

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## Example 7 – Solving a System of linear Equations (2 of 2)

Using Gaussian elimination, you can show that this system has infinitely many solutions, one of which is $x_1 = 1$, $x_2 = 1$, $x_3 = -1$.

$$1\begin{bmatrix}1\\4\\7\end{bmatrix} + 1\begin{bmatrix}2\\5\\8\end{bmatrix} + (-1)\begin{bmatrix}3\\6\\9\end{bmatrix} = \begin{bmatrix}0\\3\\6\end{bmatrix}$$

That is, **b** can be expressed as a linear combination of the columns of *A*. This representation of one column vector in terms of others is a fundamental theme of linear algebra.

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# 2 Matrices




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## 2.2 Properties of Matrix Operations



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## Objectives

- Use the properties of matrix addition, scalar multiplication, and zero matrices.
- Use the properties of matrix multiplication and the identity matrix.
- Find the transpose of a matrix.

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## Algebra of Matrices

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## Algebra of Matrices (1 of 3)

This section begins to develop the **algebra of matrices.** Theorem 2.1 lists several properties of matrix addition and scalar multiplication.

**Theorem 2.1 Properties of Matrix Addition and Scalar Multiplication**

If $A$, $B$, and $C$ are $m \times n$ matrices, and $c$ and $d$ are scalars, then the properties below are true.

**1.** $A + B = B + A$ Commutative property of addition

**2.** $A + (B + C) = (A + B) + C$ Associative property of addition

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## Algebra of Matrices (2 of 3)

**Theorem 2.1 Properties of Matrix Addition and Scalar Multiplication**

**3.** $(cd)A = c(dA)$ — Associative property of multiplication

**4.** $1A = A$ — Multiplicative identity

**5.** $c(A + B) = cA + cB$ — Distributive property

**6.** $(c + d)A = cA + dA$ — Distributive property

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## Example 1 – Addition of More than Two Matrices

To obtain the sum of four matrices, add corresponding entries as shown below.

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

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## Algebra of Matrices (3 of 3)

**Theorem 2.2 properties of zero matrices**

If $A$ is an $m \times n$ matrix and $c$ is a scalar, then the properties below are true.

**1.** $A + O_{mn} = A$

**2.** $A + (-A) = O_{mn}$

**3.** If $cA = O_{mn}$, then $c = 0$ or $A = O_{mn}$.

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## Example 2 – Solving a Matrix equation

Solve for $X$ in the equation $3X + A = B$, where

$$A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix}.$$

**Solution:**

Begin by solving the equation for $X$ to obtain

$$3X = B - A$$
$$X = \tfrac{1}{3}(B - A).$$

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## Example 2 – Solution

Now, using the matrices *A* and *B*, you have

$$X = \frac{1}{3}\left(\begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}\right)$$

$$= \frac{1}{3}\begin{bmatrix} -4 & 6 \\ 2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{4}{3} & 2 \\ \frac{2}{3} & -\frac{2}{3} \end{bmatrix}.$$

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## Properties of Matrix Multiplication

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## Properties of Matrix Multiplication (1 of 3)

**Theorem 2.3 Properties of Matrix Multiplication**

If $A$, $B$, and $C$ are matrices (with sizes such that the matrix products are defined), and $c$ is a scalar, then the properties below are true.

**1.** $A(BC) = (AB)C$ Associative property of multiplication

**2.** $A(B + C) = AB + AC$ Distributive property

**3.** $(A + B)C = AC + BC$ Distributive property

**4.** $c(AB) = (cA)B = A(cB)$

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## Example 3 – Matrix Multiplication Is Associative

Find the matrix product $ABC$ by grouping the factors first as $(AB)C$ and then as $A(BC)$.

Show that you obtain the same result from both processes.

$$A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

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# Example 3 – Solution (1 of 2)

Grouping the factors as (*AB*)*C*, you have

$$(AB)C = \left(\begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix}\right)\begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 4 & 0 \\ -1 & 2 & 3 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix}.$$



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# Example 3 – Solution (2 of 2)

Grouping the factors as *A*(*BC*), you obtain the same result.

$$A(BC) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}\left(\begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 3 & 8 \\ -7 & 2 \end{bmatrix} = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix}$$



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## Properties of Matrix Multiplication (2 of 3)

**Theorem 2.4 Properties of the Identity Matrix**

If $A$ is a matrix of size $m \times n$, then the properties below are true.

**1.** $AI_n = A$ **2.** $I_mA = A$



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## Example 6 – Multiplication by an Identity Matrix

**a.** $\begin{bmatrix} 3 & -2 \\ 4 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 4 & 0 \\ -1 & 1 \end{bmatrix}$

**b.** $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$



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## Properties of Matrix Multiplication (3 of 3)

**Theorem 2.5 Number of Solutions of a Linear System**

For a system of linear equations, precisely one of the statements below is true.

**1.** The system has exactly one solution.

**2.** The system has infinitely many solutions.

**3.** The system has no solution.

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## The Transpose of a Matrix

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## The Transpose of a Matrix (1 of 2)

The **transpose** of a matrix is formed by writing its rows as columns.





## Example 8 – The Transpose of a Matrix

Find the transpose of each matrix.

**a.** $A = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$ **b.** $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ **c.** $C = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **d.** $D = \begin{bmatrix} 0 & 1 \\ 2 & 4 \\ 1 & -1 \end{bmatrix}$

**Solution:**

**a.** $A^T = \begin{bmatrix} 2 & 8 \end{bmatrix}$ **b.** $B^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

# Example 8 – Solution

**c.** $C^T = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **d.** $D^T = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & -1 \end{bmatrix}$

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# The Transpose of a Matrix (2 of 2)

**Theorem 2.6 Properties of Transposes**

If *A* and *B* are matrices (with sizes such that the matrix operations are defined) and *c* is a scalar, then the properties below are true.

**1.** $(A^T)^T = A$ Transpose of a transpose
**2.** $(A + B)^T = A^T + B^T$ Transpose of a sum
**3.** $(cA)^T = c(A^T)$ Transpose of a scalar multiple
**4.** $(AB)^T = B^T A^T$ Transpose of a product

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# 2 Matrices






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## 2.3 The Inverse of a Matrix



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## Objectives

- Find the inverse of a matrix (if it exists).
- Use properties of inverse matrices.
- Use an inverse matrix to solve a system of linear equations.

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## Matrices and Their Inverses

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## The Inverse of a Matrix (1 of 5)

**Definition of the Inverse of a Matrix**

An $n \times n$ matrix $A$ is **invertible** (or **nonsingular**) when there exists an $n \times n$ matrix $B$ such that

$AB = BA = I_n$

where $I_n$ is the identity matrix of order $n$. The matrix $B$ is the (multiplicative) **inverse** of $A$. A matrix that does not have an inverse is **noninvertible** (or **singular**).

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## The Inverse of a Matrix (2 of 5)

**Theorem 2.7 Uniqueness of an Inverse Matrix**

If $A$ is an invertible matrix, then its inverse is unique. The inverse of $A$ is denoted by $A^{-1}$.

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## Example 2 – Finding the Inverse of a Matrix

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}.$$

**Solution:**

To find the inverse of *A*, solve the matrix equation *AX* = *I* for *X*.

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_{11} + 4x_{21} & x_{12} + 4x_{22} \\ -x_{11} - 3x_{21} & -x_{12} - 3x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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## Example 2 – Solution

Equating corresponding entries, you obtain two systems of linear equations.

$$\begin{aligned} x_{11} + 4x_{21} &= 1 \\ -x_{11} - 3x_{21} &= 0 \end{aligned} \qquad \begin{aligned} x_{12} + 4x_{22} &= 0 \\ -x_{12} - 3x_{22} &= 1 \end{aligned}$$

Solving the first system, you find that $x_{11} = -3$ and $x_{21} = 1$. Similarly, solving the second system, you find that $x_{12} = -4$ and $x_{22} = 1$. So, the inverse of *A* is

$$X = A^{-1} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}.$$

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## The Inverse of a Matrix (3 of 5)

**Finding the Inverse of a Matrix by Gauss-Jordan Elimination**

Let $A$ be a square matrix of order $n$.

1. Write the $n \times 2n$ matrix that consists of $A$ on the left and the $n \times n$ identity matrix $I$ on the right to obtain $[A \quad I]$. This process is called **adjoining** matrix $I$ to matrix $A$.
2. If possible, row reduce $A$ to $I$ using elementary row operations on the entire matrix $[A \quad I]$.

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## The Inverse of a Matrix (4 of 5)

**Finding the Inverse of a Matrix by Gauss-Jordan Elimination**

The result will be the matrix $[I \quad A^{-1}]$. If this is not possible, then $A$ is noninvertible (or singular).

3. Check your work by multiplying to see that $AA^{-1} = I = A^{-1}A$.

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## Example 3 – Finding the Inverse of a Matrix

Find the inverse of the matrix.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$

**Solution:**

Begin by adjoining the identity matrix to *A* to form the matrix

$$[A \quad I] = \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{bmatrix}.$$

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## Example 3 – Solution (1 of 3)

Use elementary row operations to obtain the form

$$[I \quad A^{-1}]$$

as shown below.

$$\begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{bmatrix} \qquad R_2 + (-1)R_1 \rightarrow R_2$$

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## Example 3 – Solution (2 of 3)

$$\left[\begin{array}{rrrrrr} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & -4 & 3 & 6 & 0 & 1 \end{array}\right] \qquad R_3 + (6)R_1 \rightarrow R_3$$

$$\left[\begin{array}{rrrrrr} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 4 & 1 \end{array}\right] \qquad R_3 + (4)R_2 \rightarrow R_3$$

$$\left[\begin{array}{rrrrrr} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{array}\right] \qquad (-1)R_3 \rightarrow R_3$$

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## Example 3 – Solution (3 of 3)

$$\left[\begin{array}{rrrrrr} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & -3 & -1 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{array}\right] \qquad R_2 + R_3 \rightarrow R_2$$

$$\left[\begin{array}{rrrrrr} 1 & 0 & 0 & -2 & -3 & -1 \\ 0 & 1 & 0 & -3 & -3 & -1 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{array}\right] \qquad R_1 + R_2 \rightarrow R_1$$

The matrix $A$ is invertible, and its inverse is

$$A^{-1} = \left[\begin{array}{rrr} -2 & -3 & -1 \\ -3 & -3 & -1 \\ -2 & -4 & -1 \end{array}\right].$$

Confirm this by showing that $AA^{-1} = I = A^{-1}A$.

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## The Inverse of a Matrix (5 of 5)

If $A$ is a 2 × 2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

then $A$ is invertible if and only if $ad - bc \neq 0$. Moreover, if $ad - bc \neq 0$, then the inverse is

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

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## Properties of Inverses

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# Properties of Inverses (1 of 3)

**Theorem 2.8 properties of inverse Matrices**

If *A* is an invertible matrix, *k* is a positive integer, and *c* is a nonzero scalar, then $A^{-1}$, $A^k$, $cA$, and $A^T$ are invertible and the statements below are true.

**1.** $(A^{-1})^{-1} = A$

**2.** $(A^k)^{-1} = \underbrace{A^{-1}A^{-1} \cdot \cdot \cdot A^{-1}}_{k \text{ factors}} = (A^{-1})^k$

**3.** $(cA)^{-1} = \frac{1}{c}A^{-1}$

**4.** $(A^T)^{-1} = (A^{-1})^T$



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# Example 6 – The Inverse of the Square of a Matrix

Compute $A^{-2}$ two different ways and show that the results are equal.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$$

**Solution:**

One way to find $A^{-2}$ is to find $(A^2)^{-1}$ by squaring the matrix *A* to obtain

$$A^2 = \begin{bmatrix} 3 & 5 \\ 10 & 18 \end{bmatrix}$$



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## Example 6 – Solution (1 of 2)

And using the formula for the inverse of a 2 × 2 matrix to obtain

$$(A^2)^{-1} = \frac{1}{4}\begin{bmatrix} 18 & -5 \\ -10 & 3 \end{bmatrix} = \begin{bmatrix} \frac{9}{2} & -\frac{5}{4} \\ -\frac{5}{2} & \frac{3}{4} \end{bmatrix}.$$

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## Example 6 – Solution (2 of 2)

Another way to find $A^{-2}$ is to find $(A^{-1})^2$ by finding $A^{-1}$

$$A^{-1} = \frac{1}{2}\begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$$

and then squaring this matrix to obtain

$$(A^{-1})^2 = \begin{bmatrix} \frac{9}{2} & -\frac{5}{4} \\ -\frac{5}{2} & \frac{3}{4} \end{bmatrix}.$$

Note that both methods produce the same result.

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## Properties of Inverses (2 of 3)

**Theorem 2.9 The Inverse of a Product**

If $A$ and $B$ are invertible matrices of order $n$, then $AB$ is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.





## Properties of Inverses (3 of 3)

**Theorem 2.10 Cancellation Properties**

If $C$ is an invertible matrix, then the properties below are true.

**1.** If $AC = BC$, then $A = B$. Right cancellation property

**2.** If $CA = CB$, then $A = B$. Left cancellation property

# Systems of Equations



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## Systems of Equations (1 of 1)

**Theorem 2.11 Systems of Equations with Unique Solutions**

If $A$ is an invertible matrix, then the system of linear equations $A\boldsymbol{x} = \mathbf{b}$ has a unique solution $\boldsymbol{x} = A^{-1}\mathbf{b}$.



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## Example 8 – Solving Systems of Equations Using an Inverse Matrix

Use an inverse matrix to solve each system.

**a.** $2x + 3y + z = -1$
$3x + 3y + z = 1$
$2x + 4y + z = -2$

**b.** $2x + 3y + z = 4$
$3x + 3y + z = 8$
$2x + 4y + z = 5$

**c.** $2x + 3y + z = 0$
$3x + 3y + z = 0$
$2x + 4y + z = 0$

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## Example 8 – Solution (1 of 3)

First note that the coefficient matrix for each system is

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix}.$$

Using Gauss-Jordan elimination,

$$A^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}.$$

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## Example 8 – Solution (2 of 3)

$$\textbf{a. } \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$$

The solution is $x = 2$, $y = -1$, and $z = -2$.

$$\textbf{b. } \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}\begin{bmatrix} 4 \\ 8 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -7 \end{bmatrix}$$

The solution is $x = 4$, $y = 1$, and $z = -7$.

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## Example 8 – Solution (3 of 3)

$$\textbf{c. } \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The solution is trivial: $x = 0$, $y = 0$, and $z = 0$.

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# 2 Matrices






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## 2.4 Elementary Matrices



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## Objectives

- Factor a matrix into a product of elementary matrices.
- Find and use an *LU*-factorization of a matrix to solve a system of linear equations.

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## Elementary Matrices and Elementary Row Operations

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## Elementary Matrices and Elementary Row Operations (1 of 6)

### Definition of an Elementary Matrix

An $n \times n$ matrix is an **elementary matrix** when it can be obtained from the identity matrix $I_n$ by a single elementary row operation.





## Elementary Matrices and Elementary Row Operations (2 of 6)

### Theorem 2.12 Representing Elementary Row Operations

Let $E$ be the elementary matrix obtained by performing an elementary row operation on $I_m$. If that same elementary row operation is performed on an $m \times n$ matrix $A$, then the resulting matrix is the product $EA$.

## Example 3 – Using Elementary Matrices

Find a sequence of elementary matrices that can be used to write the matrix *A* in row-echelon form.

$$A = \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix}$$

**Solution:**

| Matrix | Elementary Row Operation | Elementary Matrix |
|---|---|---|
| $\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 2 & -6 & 2 & 0 \end{bmatrix}$ | $R_1 \leftrightarrow R_2$ | $E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |





## Example 3 – Solution (1 of 2)

| | | |
|---|---|---|
| $\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & -4 \end{bmatrix}$ | $R_3 + (-2)R_1 \rightarrow R_3$ | $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$ |
| $\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$ | $\left(\frac{1}{2}\right)R_3 \rightarrow R_3$ | $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$ |

## Example 3 – Solution (2 of 2)

The three elementary matrices $E_1$, $E_2$, and $E_3$ can be used to perform the same elimination.

$$B = E_3E_2E_1A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 2 & -6 & 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & -4 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

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## Elementary Matrices and Elementary Row Operations (3 of 6)

**Definition of Row Equivalence**

Let $A$ and $B$ be $m \times n$ matrices. Matrix $B$ is **row-equivalent** to $A$ when there exists a finite number of elementary matrices $E_1, E_2, \ldots, E_k$ such that

$$B = E_kE_{k-1} \ldots E_2E_1A.$$

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## Elementary Matrices and Elementary Row Operations (4 of 6)

**Theorem 2.13 Elementary Matrices Are Invertible**

If $E$ is an elementary matrix, then $E^{-1}$ exists and is an elementary matrix.





## Elementary Matrices and Elementary Row Operations (5 of 6)

**Theorem 2.14 A Property of Invertible Matrices**

A square matrix $A$ is invertible if and only if it can be written as the product of elementary matrices.

## Example 4 – Writing a Matrix as the Product of Elementary Matrices

Find a sequence of elementary matrices whose product is the nonsingular matrix

$$A = \begin{bmatrix} -1 & -2 \\ 3 & 8 \end{bmatrix}.$$



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## Example 4 – Solution (1 of 2)

Begin by finding a sequence of elementary row operations that can be used to rewrite *A* in reduced row-echelon form.

| Matrix | Elementary Row Operation | Elementary Matrix |
|---|---|---|
| $\begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$ | $(-1)R_1 \rightarrow R_1$ | $E_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ |
| $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ | $R_2 + (-3)R_1 \rightarrow R_2$ | $E_2 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ |
| $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ | $(\frac{1}{2})R_2 \rightarrow R_2$ | $E_3 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ |
| $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ | $R_1 + (-2)R_2 \rightarrow R_1$ | $E_4 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$ |



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# Example 4 – Solution (2 of 2)

Now, from the matrix product $E_4E_3E_2E_1A = I$, solve for $A$ to obtain $A = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}$. This implies that $A$ is a product of elementary matrices.

$$A = \underbrace{\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}}_{E_1^{-1}} \underbrace{\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}}_{E_2^{-1}} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}}_{E_3^{-1}} \underbrace{\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}}_{E_4^{-1}} = \begin{bmatrix} -1 & -2 \\ 3 & 8 \end{bmatrix}$$

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## Elementary Matrices and Elementary Row Operations (6 of 6)

**Theorem 2.15 Equivalent Conditions**

If $A$ is an $n \times n$ matrix, then the statements below are equivalent.

**1.** $A$ is invertible.

**2.** $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ column matrix $\mathbf{b}$.

**3.** $A\mathbf{x} = O$ has only the trivial solution.

**4.** $A$ is row-equivalent to $I_n$.

**5.** $A$ can be written as the product of elementary matrices.

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# The *LU*-Factorization

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## The *LU*-Factorization (1 of 1)

**Definition of *LU*-Factorization**

If the $n \times n$ matrix $A$ can be written as the product of a lower triangular matrix $L$ and an upper triangular matrix $U$, then $A = LU$ is an ***LU*-factorization** of $A$.

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## Example 6 – Finding an *LU*-Factorization of a Matrix

Find an *LU*-factorization of the matrix $A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix}.$

**Solution:**

Begin by row reducing *A* to upper triangular form while keeping track of the elementary matrices used for each row operation.

| Matrix | Elementary Row Operation | Elementary Matrix |
|---|---|---|
| $\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & -4 & 2 \end{bmatrix}$ | $R_3 + (-2)R_1 \to R_3$ | $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$ |
| $\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$ | $R_3 + (4)R_2 \to R_3$ | $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}$ |

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## Example 6 – Solution

The reduced matrix above is an upper triangular matrix *U*, and it follows that $E_2E_1A = U$, or $A = E_1^{-1}E_2^{-1}U$. The product of the lower triangular matrices

$$E_1^{-1}E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix}$$

is a lower triangular matrix *L*, so the factorization *A* = *LU* is complete.

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## Example 7 – Solving a Linear System Using *LU*-Factorization

Solve the linear system.

$$\begin{aligned} x_1 - 3x_2 \phantom{+ 2x_3} &= -5 \\ x_2 + 3x_3 &= -1 \\ 2x_1 - 10x_2 + 2x_3 &= -20 \end{aligned}$$

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## Example 7 – Solution (1 of 4)

**Solution:**

An *LU*-factorization of the coefficient matrix *A* is given as,

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$$

First, let $\mathbf{y} = U\mathbf{x}$ and solve the system $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$.

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## Example 7 – Solution (2 of 4)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ -20 \end{bmatrix}$$

Solve this system using forward substitution. Starting with the first equation, you have $y_1 = -5$. The second equation gives $y_2 = -1$. Finally, from the third equation,

$$2y_1 - 4y_2 + y_3 = -20$$

$$y_3 = -20 - 2y_1 + 4y_2$$

$$y_3 = -20 - 2(-5) + 4(-1)$$

$$y_3 = -14.$$



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## Example 7 – Solution (3 of 4)

The solution of $L\mathbf{y} = \mathbf{b}$ is

$$\mathbf{y} = \begin{bmatrix} -5 \\ -1 \\ -14 \end{bmatrix}.$$

Now solve the system $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ using back-substitution.

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ -14 \end{bmatrix}$$



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## Example 7 – Solution (4 of 4)

From the bottom equation, $x_3 = -1$. Then, the second equation gives

$$x_2 + 3(-1) = -1$$

or $x_2 = 2$. Finally, the first equation gives

$$x_1 - 3(2) = -5$$

or $x_1 = 1$. So, the solution of the original system of equations is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}.$$

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# 2 Matrices




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## 2.5 Markov Chains



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## Objectives

- Use a stochastic matrix to find the *n*th state matrix of a Markov chain.
- Find the steady state matrix of a Markov chain.
- Find the steady state matrix of an absorbing Markov chain.

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# Stochastic Matrices and Markov Chains

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## Stochastic Matrices and Markov Chains (1 of 3)

The probability that a member of a population will change from the $j$th state to the $i$th state is represented by a number $p_{ij}$, where $0 \le p_{ij} \le 1$.

A probability of $p_{ij} = 0$ means that the member is certain *not* to change from the $j$th state to the $i$th state, whereas a probability of $p_{ij} = 1$ means that the member is certain to change from the $j$th state to the $i$th state.

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## Stochastic Matrices and Markov Chains (2 of 3)

$$
\begin{array}{cc}
 & \overbrace{\begin{array}{cccc} S_1 & S_2 & \cdots & S_n \end{array}}^{\text{From}} \\
P = & \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix} \left.\begin{array}{c} S_1 \\ S_2 \\ \vdots \\ S_n \end{array}\right\} \text{To}
\end{array}
$$

$P$ is called the **matrix of transition probabilities** because it gives the probabilities of each possible type of transition (or change) within the population.

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## Stochastic Matrices and Markov Chains (3 of 3)

At each transition, each member in a given state must either stay in that state or change to another state. For probabilities, this means that the sum of the entries in any column of $P$ is 1. For instance, in the first column,

$$p_{11} + p_{21} + \ldots + p_{n1} = 1.$$

Such a matrix is called **stochastic** (the term “stochastic” means “regarding conjecture”). That is, an $n \times n$ matrix $P$ is a **stochastic matrix** when each entry is a number between 0 and 1 inclusive, and the sum of the entries in each column of $P$ is 1.

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## Example 1 – Examples of Stochastic Matrices and Nonstochastic Matrices

The matrices in parts (a), (b), and (c) are stochastic, but the matrices in parts (d), (e), and (f) are not.

**a.** $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ **b.** $\begin{bmatrix} \frac{1}{4} & \frac{1}{5} & \frac{1}{3} & \frac{1}{2} \\ \frac{1}{4} & \frac{13}{60} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{3} & \frac{1}{6} \end{bmatrix}$ **c.** $\begin{bmatrix} 0.9 & 0.8 \\ 0.1 & 0.2 \end{bmatrix}$

**d.** $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} \end{bmatrix}$ **e.** $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{4} & \frac{3}{4} & 0 \end{bmatrix}$ **f.** $\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \\ 0.2 & 0.3 & 0.4 & 0.5 \\ 0.3 & 0.4 & 0.5 & 0.6 \\ 0.4 & 0.5 & 0.6 & 0.7 \end{bmatrix}$

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## Stochastic Matrices and Markov Chains (4 of 3)

### *n*th State Matrix of a Markov Chain

The *n*th state matrix of a Markov chain for which $P$ is the matrix of transition probabilities and $X_0$ is the initial state matrix is

$$X_n = P^n X_0.$$

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## Example 3 – A Consumer Preference Model

Assuming the given matrix of transition probabilities remains the same year after year,

$$P = \begin{array}{c} \\ \\ \end{array} \overbrace{\begin{array}{ccc} \text{A} & \text{B} & \text{None} \end{array}}^{\text{From}}$$

$$P = \begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix} \left.\begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array}\right\} \text{To}$$

find the number of subscribers each satellite television company will have after (a) 3 years, (b) 5 years, (c) 10 years, and (d) 15 years.

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## Example 3 – Solution (1 of 4)

**a.** To find the numbers of subscribers after 3 years, first find $X_3$.

$$X_3 = P^3X_0 \approx \begin{bmatrix} 0.3028 \\ 0.3904 \\ 0.3068 \end{bmatrix} \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 3 years}$$

After 3 years, Company A will have about

0.3028(100,000) = 30,280 subscribers and Company B will have about

0.3904(100,000) = 39,040 subscribers.

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## Example 3 – Solution (2 of 4)

**b.** To find the numbers of subscribers after 5 years, first find $X_5$.

$$X_5 = P^5 X_0 \approx \begin{bmatrix} 0.3241 \\ 0.4381 \\ 0.2378 \end{bmatrix} \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 5 years}$$

After 5 years, Company A will have about 0.3241(100,000) = 32,410 subscribers and Company B will have about 0.4381(100,000) = 43,810 subscribers.

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## Example 3 – Solution (3 of 4)

**c.** To find the numbers of subscribers after 10 years, first find $X_{10}$.

$$X_{10} = P^{10} X_0 \approx \begin{bmatrix} 0.3329 \\ 0.4715 \\ 0.1957 \end{bmatrix} \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 10 years}$$

After 10 years, Company A will have about 0.3329(100,000) = 33,290 subscribers and Company B will have about 0.4715(100,000) = 47,150 subscribers.

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## Example 3 – Solution (4 of 4)

**d.** To find the numbers of subscribers after 15 years, first find $X_{15}$.

$$X_{15} = P^{15}X_0 \approx \begin{array}{l} \begin{bmatrix} 0.3333 \\ 0.4756 \\ 0.1911 \end{bmatrix} \end{array} \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 15 years}$$

After 15 years, Company A will have about 0.3333(100,000) = 33,330 subscribers and Company B will have about 0.4756(100,000) = 47,560 subscribers.

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## Steady State Matrix of a Markov Chain

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## Example 4 – Regular Stochastic Matrices (1 of 3)

**a.** The stochastic matrix

$$P = \begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix}$$

is regular because $P^1$ has only positive entries.



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## Example 4 – Regular Stochastic Matrices (2 of 3)

**b.** The stochastic matrix

$$P = \begin{bmatrix} 0.50 & 1.00 \\ 0.50 & 0 \end{bmatrix}$$

is regular because

$$P^2 = \begin{bmatrix} 0.75 & 0.50 \\ 0.25 & 0.50 \end{bmatrix}$$

has only positive entries.



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## Example 4 – Regular Stochastic Matrices (3 of 3)

**c.** The stochastic matrix

$$P = \begin{bmatrix} \frac{1}{3} & 0 & 1 \\ \frac{1}{3} & 1 & 0 \\ \frac{1}{3} & 0 & 0 \end{bmatrix}$$

is not regular because every power of $P$ has two zeros in its second column.

When $P$ is a regular stochastic matrix, the corresponding **regular Markov chain** $PX_0, P^2X_0, P^3X_0, \ldots$ approaches a unique **steady state matrix** $\overline{X}$.

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## Example 5 – Finding a Steady State Matrix

Find the steady state matrix $\overline{X}$ of the Markov chain whose matrix of transition probabilities is the regular matrix

$$P = \begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix}.$$

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## Example 5 – Solution (1 of 3)

Note that $P$ is the matrix of transition probabilities with steady state matrix $\overline{X}$. To *find* $\overline{X}$, begin by

letting $\overline{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Then use the matrix equation $P\overline{X} = \overline{X}$ to obtain

$$\begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Or

$$0.70x_1 + 0.15x_2 + 0.15x_3 = x_1$$
$$0.20x_1 + 0.80x_2 + 0.15x_3 = x_2$$
$$0.10x_1 + 0.05x_2 + 0.70x_3 = x_3.$$

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## Example 5 – Solution (2 of 3)

Use these equations and the fact that $x_1 + x_2 + x_3 = 1$ to write the system of linear equations below.

$$-0.30x_1 + 0.15x_2 + 0.15x_3 = 0$$
$$0.20x_1 - 0.20x_2 + 0.15x_3 = 0$$
$$0.10x_1 + 0.05x_2 - 0.30x_3 = 0$$
$$x_1 + x_2 + x_3 = 1.$$

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# Example 5 – Solution (3 of 3)

Use any appropriate method to verify that the solution of this system is

$$x_1 = \tfrac{1}{3}, \quad x_2 = \tfrac{10}{21}, \quad \text{and} \quad x_3 = \tfrac{4}{21}.$$

So the steady state matrix is

$$\overline{X} = \begin{bmatrix} \frac{1}{3} \\ \frac{10}{21} \\ \frac{4}{21} \end{bmatrix} \approx \begin{bmatrix} 0.3333 \\ 0.4762 \\ 0.1905 \end{bmatrix}.$$

Check that $P\overline{X} = \overline{X}$.

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# Steady State Matrix of a Markov Chain (1 of 1)

A summary for finding the steady state matrix $\overline{X}$ of a Markov chain is below.

**Finding the Steady State Matrix of a Markov chain**

**1.** Check to see that the matrix of transition probabilities $P$ is a regular matrix.

**2.** Solve the system of linear equations obtained from the matrix equation $P\overline{X} = \overline{X}$ along with the equation $x_1 + x_2 + . . . + x_n = 1$.

**3.** Check the solution found in Step 2 in the matrix equation $P\overline{X} = \overline{X}$.

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# Absorbing Markov Chains

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## Absorbing Markov Chains

Consider a Markov chain with $n$ different states

$\{S_1, S_2, \ldots, S_n\}$. The $i$th *state* $S_i$ is an **absorbing state** when, in the matrix of transition probabilities $P$, $p_{ii} = 1$. That is, the entry on the main diagonal of $P$ is 1 and all other entries in the $i$th column of $P$ are 0. An **absorbing Markov chain** has the two properties listed below.

**1.** The Markov chain has at least one absorbing state.

**2.** It is possible for a member of the population to move from any nonabsorbing state to an absorbing state in a finite number of transitions.

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## Example 7 – Finding Steady State Matrices of Absorbing Markov Chains

Find the steady state matrix $\overline{X}$ of each absorbing Markov chain with matrix of transition probabilities *P*.

**a.** $P = \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 1 & 0.5 \\ 0.6 & 0 & 0.5 \end{bmatrix}$ **b.** $P = \begin{bmatrix} 0.5 & 0 & 0.2 & 0 \\ 0.2 & 1 & 0.3 & 0 \\ 0.1 & 0 & 0.4 & 0 \\ 0.2 & 0 & 0.1 & 1 \end{bmatrix}$

**Solution:**

**a.** Use the matrix equation $P\overline{X} = \overline{X}$, or

$$\begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 1 & 0.5 \\ 0.6 & 0 & 0.5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

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## Example 7 – Solution (1 of 3)

Along with the equation $x_1 + x_2 + x_3 = 1$ to write the system of linear equations

$$\begin{aligned} -0.6x_1 \qquad\qquad &= 0 \\ 0.5x_3 &= 0 \\ 0.6x_1 \quad - 0.5x_3 &= 0 \\ x_1 + x_2 + \quad x_3 &= 1. \end{aligned}$$

The solution of this system is $x_1 = 0$, $x_2 = 1$, and $x_3 = 0$, so the steady state matrix is $\overline{X} = [0 \quad 1 \quad 0]^T$. Note that $\overline{X}$ coincides with the second column of the matrix of transition probabilities *P*.

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## Example 7 – Solution (2 of 3)

**b.** Use the matrix equation $P\overline{X} = \overline{X}$, or

$$\begin{bmatrix} 0.5 & 0 & 0.2 & 0 \\ 0.2 & 1 & 0.3 & 0 \\ 0.1 & 0 & 0.4 & 0 \\ 0.2 & 0 & 0.1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

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## Example 7 – Solution (3 of 3)

Along with the equation $x_1 + x_2 + x_3 + x_4 = 1$ to write the system of linear equations

$$\begin{aligned} -0.5x_1 + \quad 0.2x_3 \quad &= 0 \\ 0.2x_1 + \quad 0.3x_3 \quad &= 0 \\ 0.1x_1 - \quad 0.6x_3 \quad &= 0 \\ 0.2x_1 + \quad 0.1x_3 \quad &= 0 \\ x_1 + x_2 + \quad x_3 + x_4 &= 1. \end{aligned}$$

The solution of this system is $x_1 = 0$, $x_2 = 1 - t$, $x_3 = 0$, and $x_4 = t$, where $t$ is any real number such that $0 \le t \le 1$. So, the steady state matrix is $\overline{X} = [0 \quad 1 - t \quad 0 \quad t]^T$. The Markov chain has an infinite number of steady state matrices.

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# 2 Matrices






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## 2.6 More Applications of Matrix Operations



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## Objectives

- Use matrix multiplication to encode and decode messages.
- Use matrix algebra to analyze an economic system (Leontief input-output model).
- Find the least squares regression line for a set of data.

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## Cryptography

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## Cryptography (1 of 2)

A **cryptogram** is a message written according to a secret code (the Greek word *kryptos* means "hidden"). One method of using matrix multiplication to **encode** and **decode** messages is introduced below.

To begin, assign a number to each letter in the alphabet (with 0 assigned to a blank space), as shown.

| | | |
|---|---|---|
| 0 = ___ | 9 = I | 18 = R |
| 1 = A | 10 = J | 19 = S |
| 2 = B | 11 = K | 20 = T |
| 3 = C | 12 = L | 21 = U |

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## Cryptography (2 of 2)

| | | |
|---|---|---|
| 4 = D | 13 = M | 22 = V |
| 5 = E | 14 = N | 23 = W |
| 6 = F | 15 = O | 24 = X |
| 7 = G | 16 = P | 25 = Y |
| 8 = H | 17 = Q | 26 = Z |

Then convert the message to numbers and partition it into **uncoded row matrices,** each having $n$ entries.

To **encode** a message, choose an $n \times n$ invertible matrix $A$ and multiply the uncoded row matrices (on the right) by $A$ to obtain **coded row matrices.**

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## Example 2 – Encoding a Message

Use the invertible matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

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## Example 2 – Solution (1 of 2)

Obtain the coded row matrices by multiplying each of the uncoded row matrices

| Uncoded Row Matrix | Encoding Matrix $A$ | Coded Row Matrix |
|---|---|---|

$$[13 \quad 5 \quad 5]\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} = [13 \quad -26 \quad 21]$$

$$[20 \quad 0 \quad 13]\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} = [33 \quad -53 \quad -12]$$

$$[5 \quad 0 \quad 13]\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} = [18 \quad -23 \quad -42]$$

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## Example 2 – Solution (2 of 2)

$$[15 \quad 14 \quad 4]\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} = [5 \quad -20 \quad 56]$$

$$[1 \quad 25 \quad 0]\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} = [-24 \quad 23 \quad 77]$$

The sequence of coded row matrices is

[13 −26 21][33 −53 −12][18 −23 −42][5 −20 56][−24 23 77].

Finally, removing the matrix notation produces the cryptogram

13 −26 21 33 −53 −12 18 −23 −42 5 −20 56 −24 23 77.

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## Example 3 – Decoding a Message

Use the inverse of the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to decode the cryptogram

13 −26 21 33 −53 −12 18 −23 −42 5 −20 56 −24 23 77.

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# Example 3 – Solution (1 of 3)

Begin by using Gauss-Jordan elimination to find $A^{-1}$.

$$\underbrace{\begin{bmatrix} 1 & -2 & 2 & 1 & 0 & 0 \\ -1 & 1 & 3 & 0 & 1 & 0 \\ 1 & -1 & -4 & 0 & 0 & 1 \end{bmatrix}}_{[A \quad I]} \rightarrow \underbrace{\begin{bmatrix} 1 & 0 & 0 & -1 & -10 & -8 \\ 0 & 1 & 0 & -1 & -6 & -5 \\ 0 & 0 & 1 & 0 & -1 & -1 \end{bmatrix}}_{[I \quad A^{-1}]}$$

Now, to decode the message, partition the message into groups of three to form the coded row matrices

[13 −26 21][33 −53 −12][18 −23 −42][5 −20 56][−24 23 77].

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# Example 3 – Solution (2 of 3)

To obtain the decoded row matrices, multiply each coded row matrix by $A^{-1}$ (on the right).

| Coded Row Matrix | Decoding Matrix $A^{-1}$ | Decoded Row Matrix |
|---|---|---|

$$[13 \quad -26 \quad 21]\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [13 \quad 5 \quad 5]$$

$$[33 \quad -53 \quad -12]\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [20 \quad 0 \quad 13]$$

$$[18 \quad -23 \quad -42]\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [5 \quad 0 \quad 13]$$

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## Example 3 – Solution (3 of 3)

$$[5 \quad -20 \quad 56]\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [15 \quad 14 \quad 4]$$

$$[-24 \quad 23 \quad 77]\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [1 \quad 25 \quad 0]$$

The sequence of decoded row matrices is

[13  5  5][20  0  13][5  0  13][15  14  4][1  25  0]

and the message is

13  5  5 20 0 13 5 0 13 15 14  4  1  25 0.

*M E E T _ M E _ M O N D A Y _*

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## Leontief Input-Output Models

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## Leontief Input-Output Models (1 of 1)

Let $d_{ij}$ be the amount of output the *j*th industry needs from the *i*th industry to produce one unit of output per year. The matrix of these coefficients is the **input-output matrix.**

$$D = \begin{array}{c} \overbrace{\begin{array}{cccc} I_1 & I_2 & \cdots & I_n \end{array}}^{\text{User (Output)}} \\ \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1n} \\ d_{21} & d_{22} & \cdots & d_{2n} \\ \vdots & \vdots & & \vdots \\ d_{n1} & d_{n2} & \cdots & d_{nn} \end{bmatrix} \end{array} \left. \begin{array}{c} I_1 \\ I_2 \\ \vdots \\ I_n \end{array} \right\} \text{Supplier (Input)}$$

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## Example 5 – Solving for the Output Matrix of an Open System

An economic system composed of three industries has the input-output matrix shown below.

$$D = \begin{array}{c} \overbrace{\begin{array}{ccc} \text{A} & \text{B} & \text{C} \end{array}}^{\text{User (Output)}} \\ \begin{bmatrix} 0.1 & 0.43 & 0 \\ 0.15 & 0 & 0.37 \\ 0.23 & 0.03 & 0.02 \end{bmatrix} \end{array} \left. \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \end{array} \right\} \text{Supplier (Input)}$$

Find the output matrix $X$ when the external demands are

$$E = \begin{bmatrix} 20{,}000 \\ 30{,}000 \\ 25{,}000 \end{bmatrix}. \quad \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \end{array}$$

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## Example 5 – Solution (1 of 2)

Letting $I$ be the identity matrix, write the equation $X = DX + E$ as $IX - DX = E$, which means that $(I - D)X = E$. Using the matrix $D$ above produces

$$I - D = \begin{bmatrix} 0.9 & -0.43 & 0 \\ -0.15 & 1 & -0.37 \\ -0.23 & -0.03 & 0.98 \end{bmatrix}.$$

Using Gauss-Jordan elimination,

$$(I - D)^{-1} \approx \begin{bmatrix} 1.25 & 0.55 & 0.21 \\ 0.30 & 1.14 & 0.43 \\ 0.30 & 0.16 & 1.08 \end{bmatrix}.$$

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## Example 5 – Solution (2 of 2)

So, the output matrix is

$$X = (I - D)^{-1}E \approx \begin{bmatrix} 1.25 & 0.55 & 0.21 \\ 0.30 & 1.14 & 0.43 \\ 0.30 & 0.16 & 1.08 \end{bmatrix} \begin{bmatrix} 20{,}000 \\ 30{,}000 \\ 25{,}000 \end{bmatrix} = \begin{array}{c} \begin{bmatrix} 46{,}750 \\ 50{,}950 \\ 37{,}800 \end{bmatrix} \end{array} \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \end{array}$$

To produce the given external demands, the outputs of the three industries must be approximately 46,750 units for industry A, 50,950 units for industry B, and 37,800 units for industry C.

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# Least Squares Regression Analysis

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## Least Squares Regression Analysis (1 of 2)

The model that has the best fit is the one that minimizes the sum of squared error. This model is the **least squares regression line,** and the procedure for finding it is the **method of least squares.**

**Definition of Least Squares Regression Line**

For a set of points

$(x_1, y_1), (x_2, y_2), \ldots, (x_n\ y_n)$

the **least squares regression line** is the linear function

$f(x) = a_0 + a_1x$

that minimizes the sum of squared error

$[y_1 - f(x_1)]^2 + [y_2 - f(x_2)]^2 + \cdot \cdot \cdot + [y_n - f(x_n)]^2.$

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## Least Squares Regression Analysis (2 of 2)

**Matrix Form for Linear Regression**

For the regression model $Y = XA + E$, the coefficients of the least squares regression line are given by the matrix equation

$$A = (X^T X)^{-1} X^T Y$$

and the sum of squared error is $E^T E$.





## Example 7 – Finding the Least Squares Regression Line

Find the least squares regression line for the points (1, 1), (2, 2), (3, 4), (4, 4), and (5, 6).

**Solution:**

The matrices $X$ and $Y$ are

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 4 \\ 6 \end{bmatrix}.$$

# Example 7 – Solution (1 of 3)

This means that

$$X^TX = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}$$

and

$$X^TY = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 63 \end{bmatrix}.$$

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# Example 7 – Solution (2 of 3)

Now, using $(X^TX)^{-1}$ to find the coefficient matrix *A*, you have

$$\begin{aligned} A &= (X^TX)^{-1}X^TY \\ &= \frac{1}{50}\begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix}\begin{bmatrix} 17 \\ 63 \end{bmatrix} \\ &= \begin{bmatrix} -0.2 \\ 1.2 \end{bmatrix}. \end{aligned}$$

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# Example 7 – Solution (3 of 3)

So, the least squares regression line is

$$y = -0.2 + 1.2x$$

as shown in Figure 2.8.

The sum of squared error for this line is 0.8 (verify this), which means that this line fits the data better than either of the two experimental linear models determined earlier.

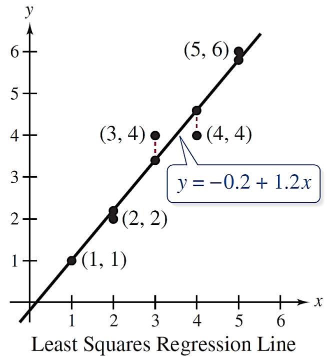


Figure 2.8



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